MC Fundamentals Task Sheet

Earthworks Excavator Design Guidance

Modes, Offsets, Points, Data, Lane Guidance

Learning Objective

To generate guidance information, the system measures the blade's position relative to a three dimensional (3D) digital map of the design surface. This is called Design guidance. 3D sensors such as GNSS receivers allow the system to always know the machine's three dimensional location

Using Design Mode

Design mode lets you select and load a design surface to get guidance to. Designs include a surface and optional line work. You can select a line and get guidance to it from the blade focus tip.

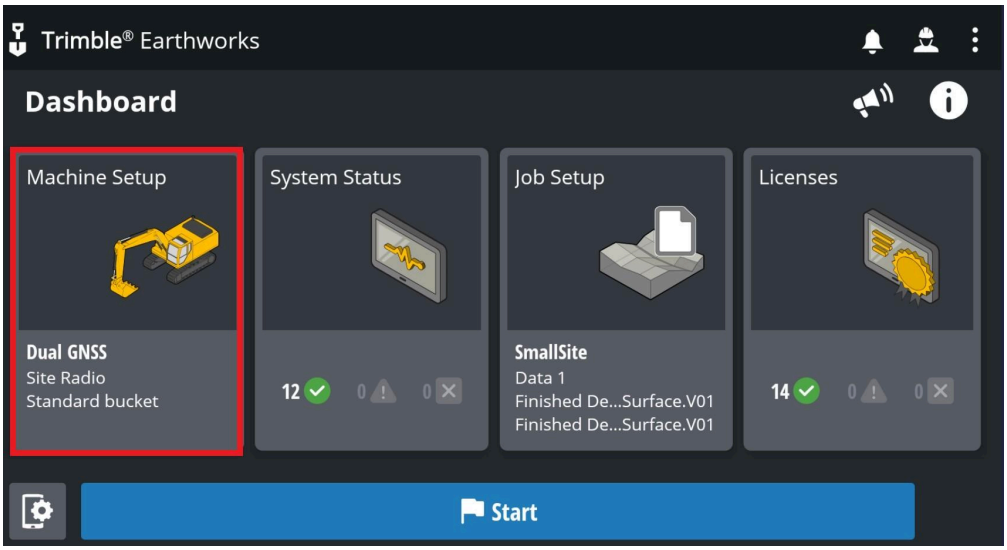
Use .dsz and .vcl design files that are created in TBC. The design files are either automatically downloaded to your system from a cloud service, or you can load them onto your system from a USB drive.

When you select a .vcl design file that contains multiple surfaces, you must further select an individual surface from the file and a master alignment for that surface.

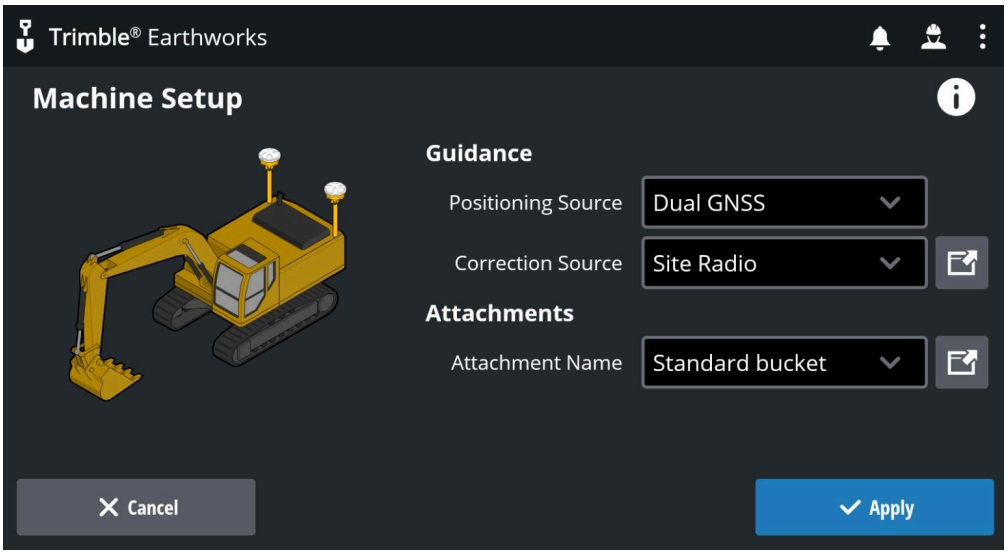
© 2018, Trimble Navigation Limited. All rights reserved. Trimble and the Globe & Triangle logo are trademarks of Trimble Navigation Limited registered in the United States and in other countries. All other trademarks are the property of their respective owners.

Selecting Design Mode

1. From the Dashboard, tap the Machine Setup tile.

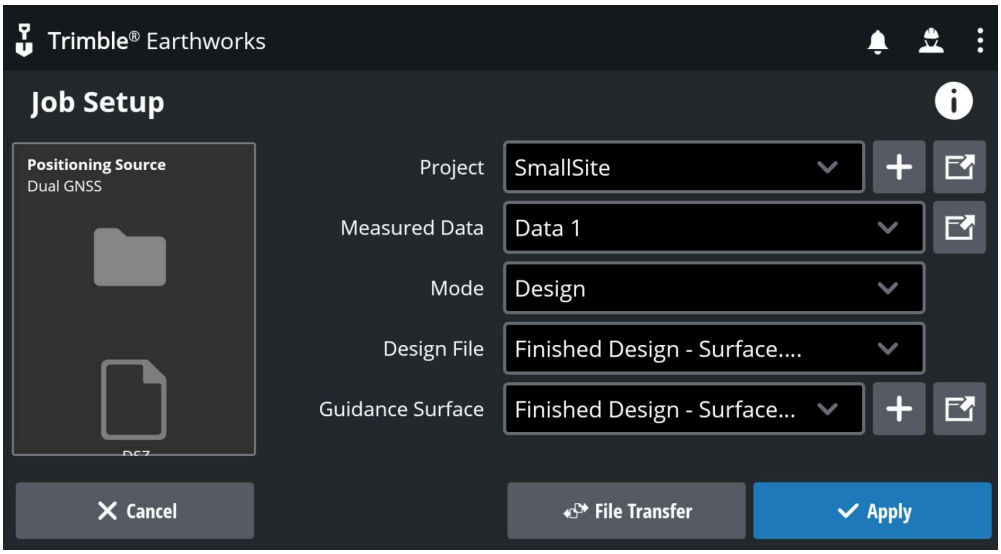


2. From the Machine Setup screen, select the Dual GNSS positioning source and tap Apply.



Note – The positioning source you select will determine the modes that you can select from the Job Setup screen.

3. From the Dashboard, tap the Job Setup tile. From the Job Setup screen select Design as the mode. Also select a project, and design file along with a surface and master alignment (if available).



Using Elevation Offsets

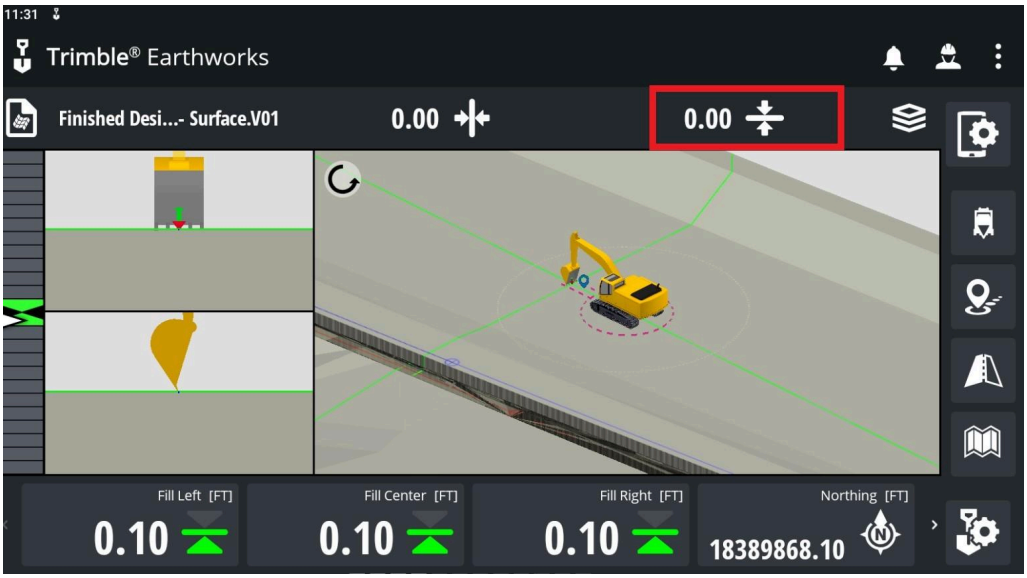
The system lets you apply elevation offsets to the design to create target surfaces that are achievable in a single pass. You can also use a working surface offset to shift the design surface to a known distance above or below the design surface, for example a subgrade at a known distance below the finished level design. When using a working surface offset, elevation offsets are applied away from the working surface so you can work the offset down to zero over several passes to reach subgrade.

1. Access the Elevation Offset screen

The Elevation offset menu is accessible via two methods. Choose one method to access this screen:

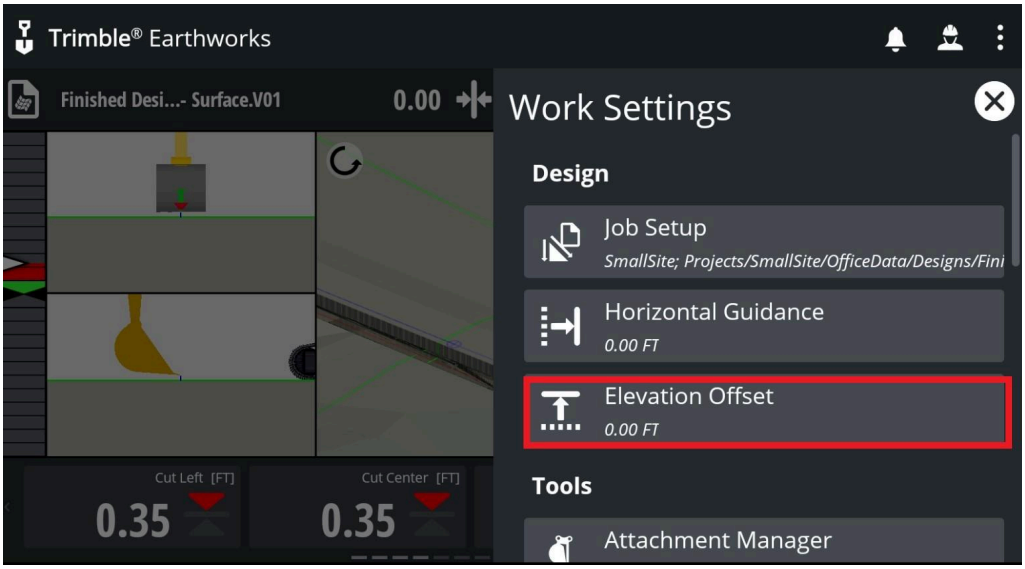
<u>Method 1</u>

Press and hold the Elevation Offset button:



Method 2

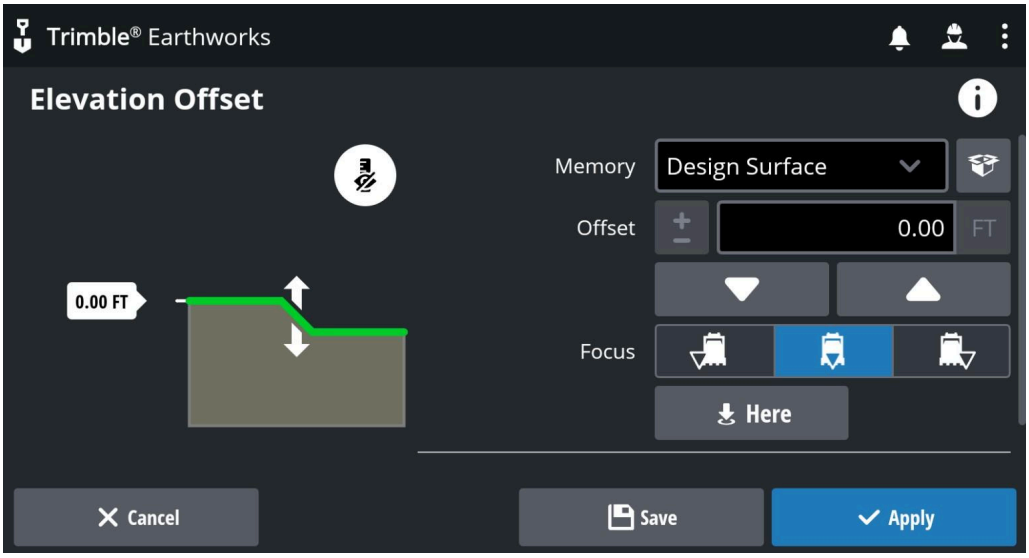
Access the Work Settings menu and choose the Elevation Offset menu:



2. Enter the following data and then tap Apply:
   - Offset value
   - Select Focus point for guidance
   - Offset direction (optional item under Advanced Options):

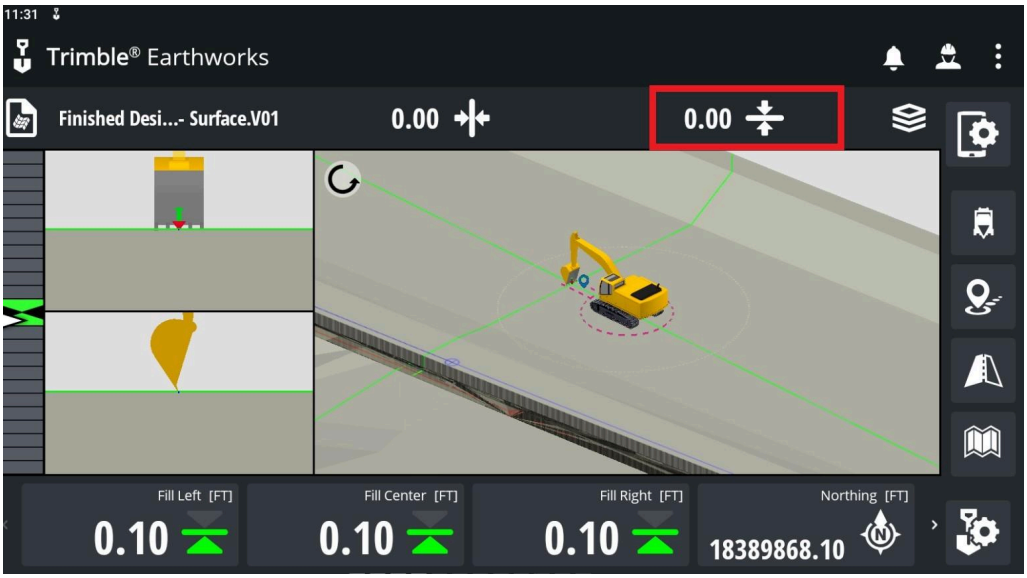
Vertical - typically for a known elevation

Perpendicular - typically for a known thickness, particularly on a sloping surface



Offset values can be saved as Memories. These are accessible to the operator via tapping on the Elevation Offset field:

3. Open the work screen and tap the elevation offset field in the guidance bar to cycle through the saved offset memories.



4. Use the increment/decrement elevation offset switch to increase or decrease the offset. (To edit the increment values, go to System Settings > Increments.)

Horizontal Guidance

You can receive horizontal guidance relative to the following design features:

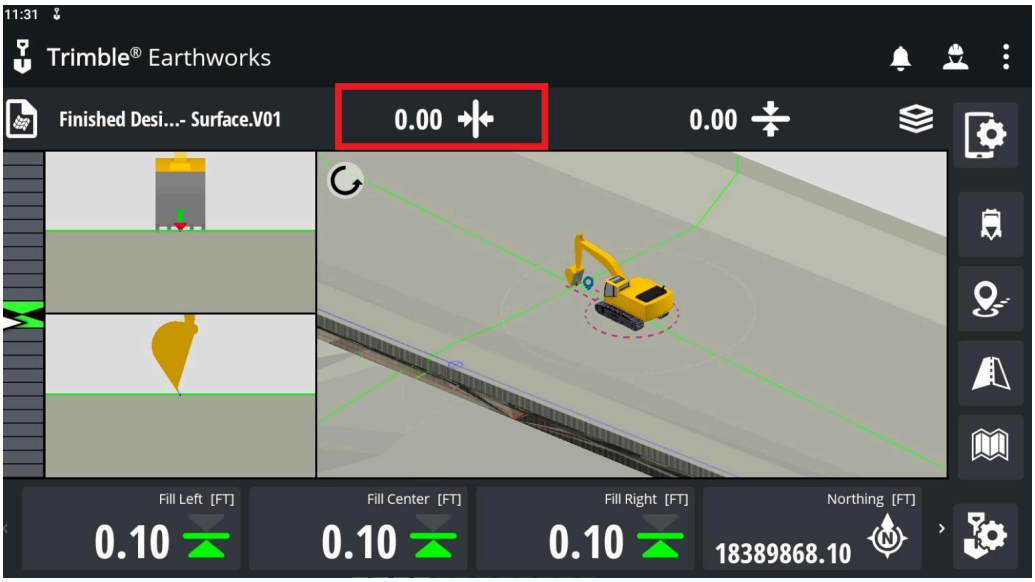
- Lines within Office Designs or Infield Designs
- Design surface boundaries
- Setting a horizontal offset moves the horizontal guidance away from the selected line. You
- Add or subtract an offset square to the selected horizontal line.

Note- Design must have linework present for this functionality to be available

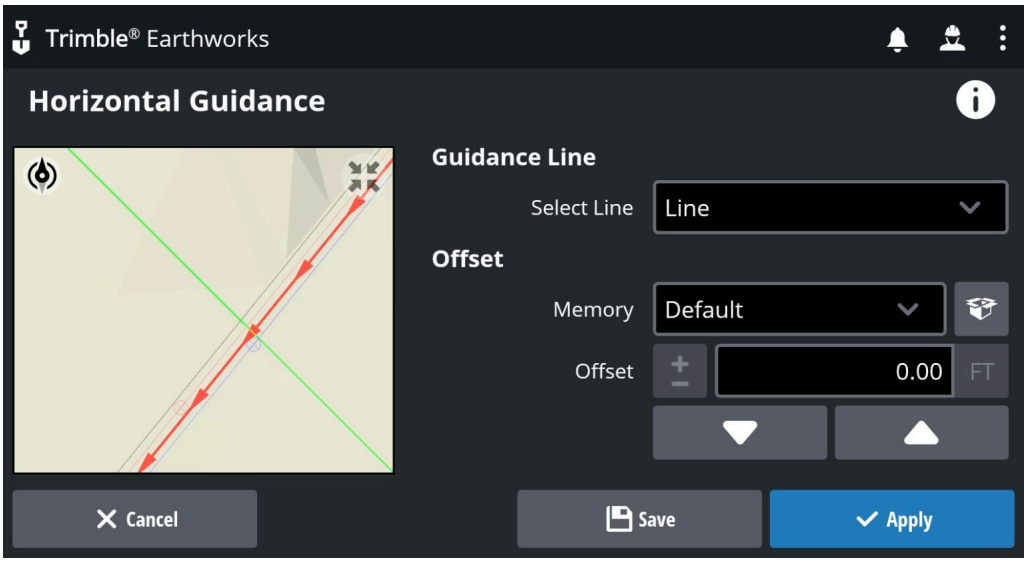
1. Access the Horizontal Guidance screen.

<u>Method 1</u>

Touch and hold the horizontal offset field in the guidance bar.



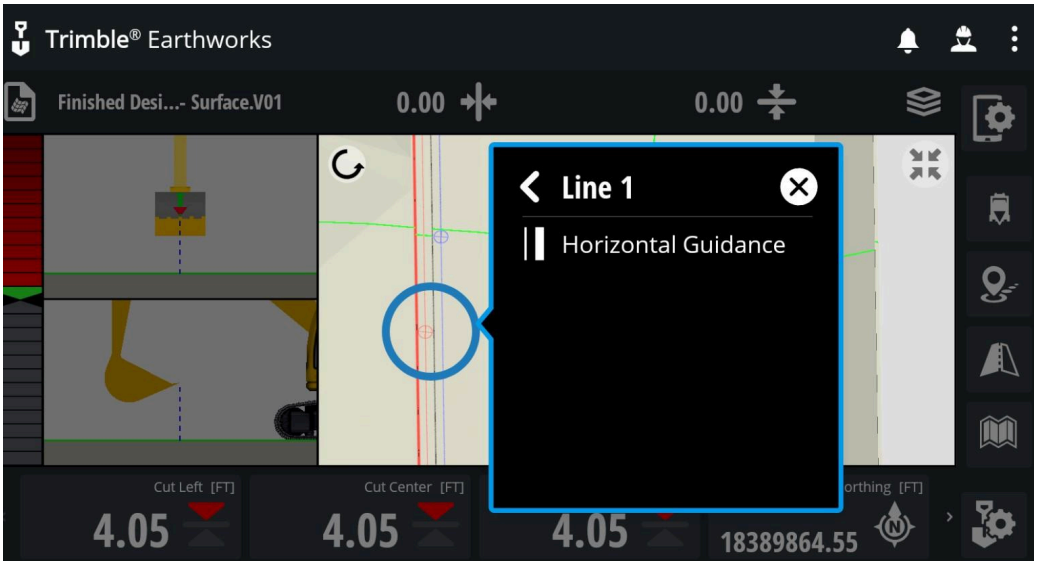
2. Select a line for horizontal guidance and select Apply.



Method 2

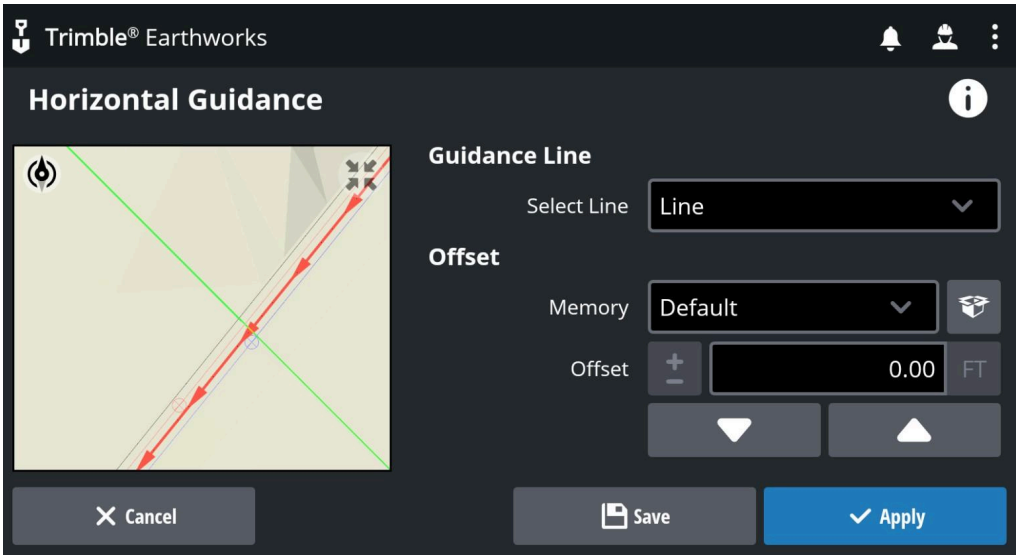
Alternatively, you can also select a line for horizontal guidance from the context menu in the work screen:

1) From the plan view guidance pane, press and hold anywhere near the line you want to select. A context menu appears.
2) Tap Select 2D Line and select an available line.
3) Tap Horizontal Guidance.

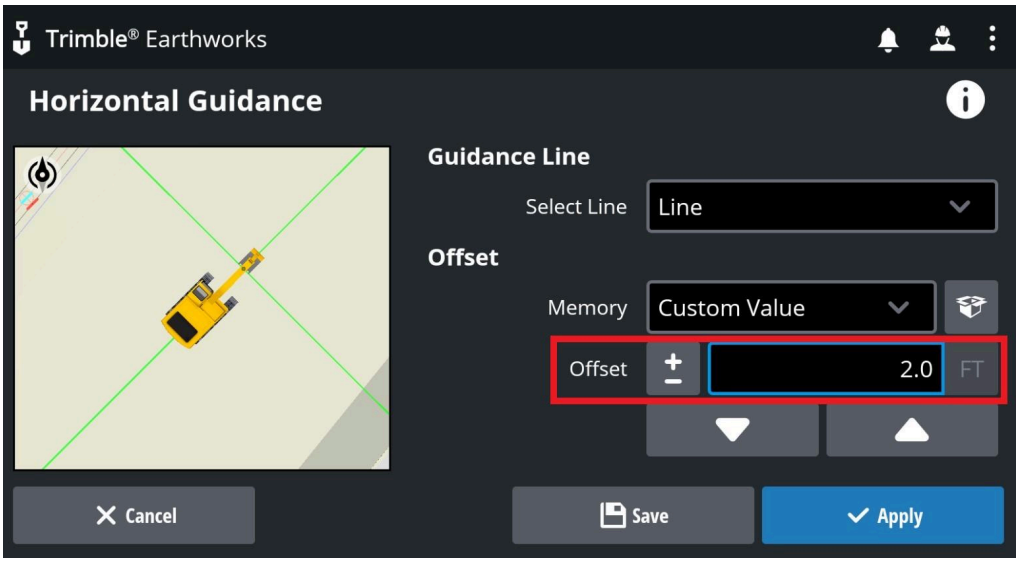


Apply a horizontal offset:

1. Once inside the Horizontal Guidance Menu, Select a line to guide to.



2. Enter the value of the offset or tap the up or down buttons to increase or decrease the value.



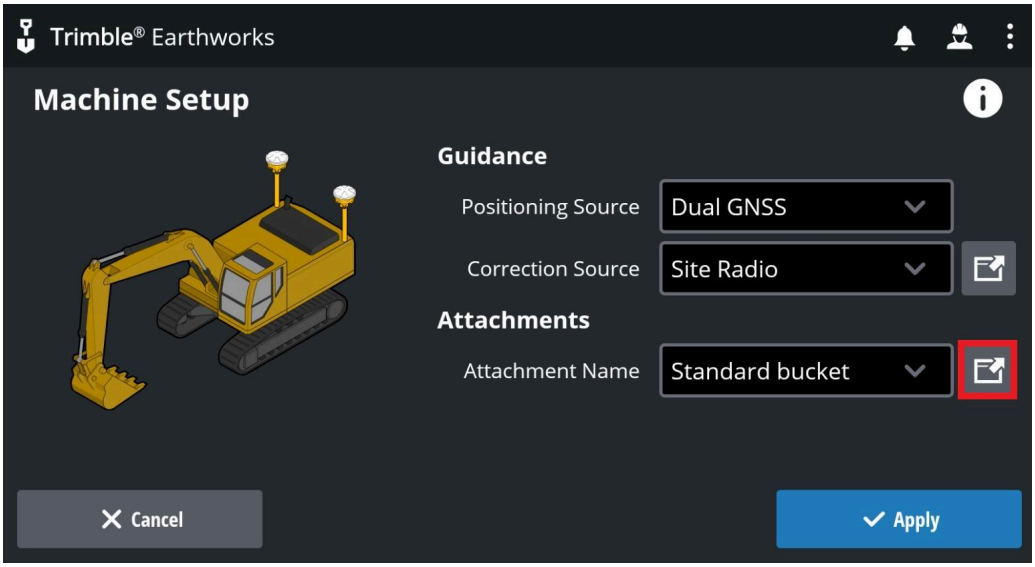
Note- To save this as a memory, tap Save.

3. Tap Apply

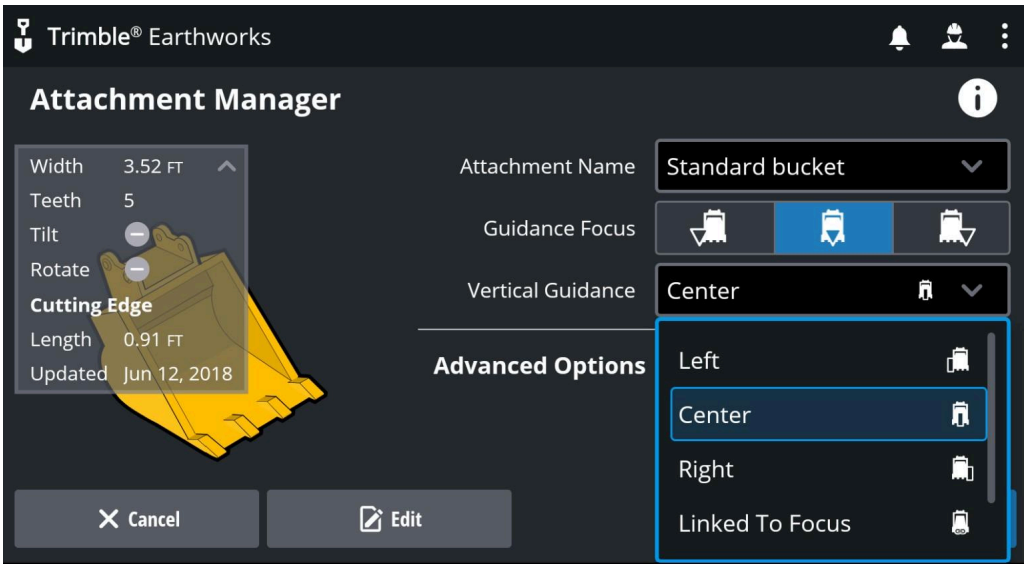
Vertical Guidance

The vertical guidance option selected in the Attachment Manager screen changes how the system looks at and extends the design surface beneath the blade to calculate the cut / fill values.

1. To select an option for Vertical Guidance access the Attachment Manager screen by going to the Machine Setup tile and selecting the menu button:



2. Select a focus, or side of the bucket or blade to guide to:



3. Select each of the guidance options available under Advanced Options below to test:

Note: some of these may be specific to certain machines and may not be available with your setup

- Right
- Center
- Left
- Linked to focus
- 2 points

4. Tap Apply

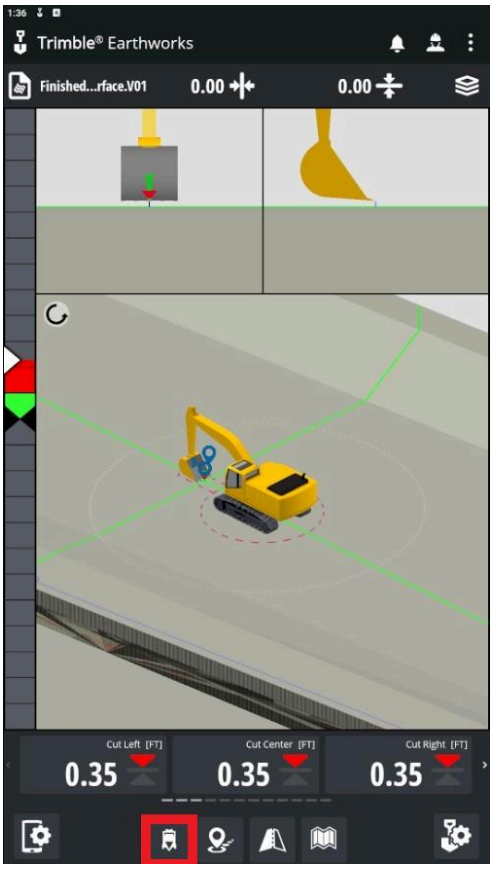
Apply a Custom Inset

The Attachment Manager screen can be accessed with 2 different methods. Choose one method from below to configure a Custom Inset:

Note- *For 2 point guidance there is no custom inset option for a bucket.*

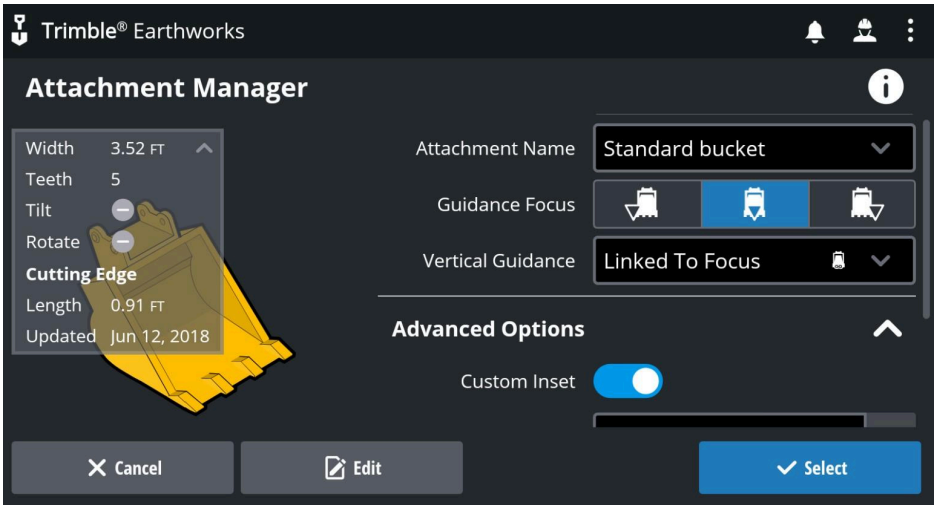
<u>Method 1</u>

Press and hold the Blade Focus shortcut button until the Attachment Manager screen appears:



Method 2

Access Blade Manager screen via the Machine Settings menu.



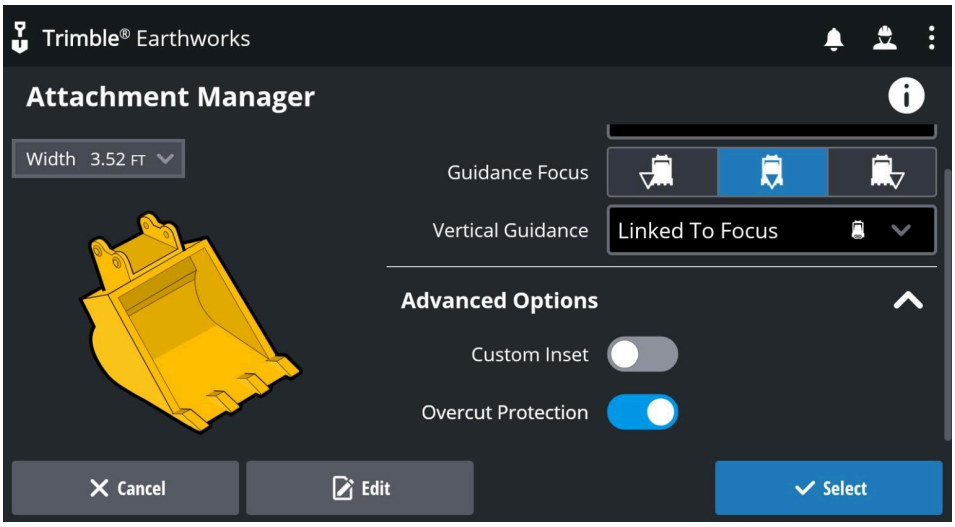
1. Select either Linked to Focus from the vertical guidance menu. Open Advanced Options and toggle on Custom Inset.

2. Enter the custom Inset value desired.

3. Tap apply.

Overcut Protection

1. Access the Attachment or Blade Manager Screen.Open Advanced Options and toggle Overcut protection on,.



2. Tap Apply

Note- Overcut Protection is useful when moving material across the toe of a slope, for example moving excess material down from the batter to the floor of the excavation.

Toggle Overcut Protection off when the task is done to receive accurate guidance on the rest of the job.

When Overcut Protection is enabled the surface adjusts to protect the toe of the slope.